Wilson-Cowan Model

In computational neuroscience, the **Wilson-Cowan model** (Hugh R. Wilson & Jack D. Cowan) describes the dynamics of interactions between populations of very simple excitatory and inhibitory model neurons. It is a mean-field model that tracks the **proportion of active neurons** in each population over time.

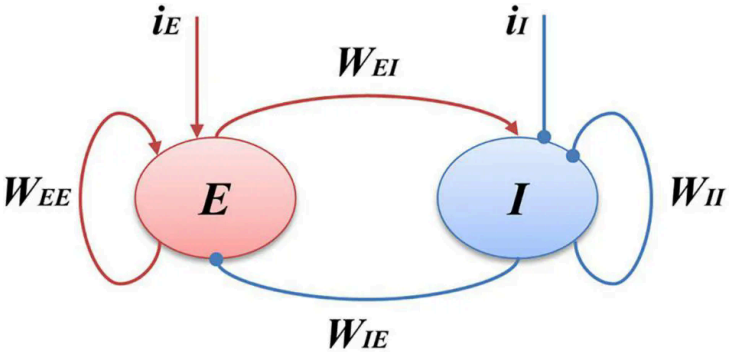

Figure 1: Schematic of the Wilson-Cowan model[1]

$$\tau_E \frac{dr_E}{dt} = -r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) \tag{1}$$

$$\tau_I \frac{dr_I}{dt} = -r_I + F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{2}$$

where the activation function F is a sigmoid:

$$F(x; a, \theta) = \frac{1}{1 + e^{-a(x-\theta)}} - \frac{1}{1 + e^{a\theta}} \tag{3}$$

The correction term $-\frac{1}{1+e^{a\theta}}$ ensures $F(0) = 0$ (no input yields no activation).

Remark: We use the simplified Wilson-Cowan equations without the refractory term $(1 - r_E)$, following the formulation in Neuromatch Academy.

0.1 Variables

- $r_E(t)$ — average firing rate (or proportion of active neurons) of the **excitatory** population at time t. Ranges from 0 (silent) to 1 (maximally active).
- $r_I(t)$ — average firing rate of the **inhibitory** population at time t.

0.2 Parameters

0.2.1 Time Constants

- τ_E — time constant of the excitatory population.
- τ_I — time constant of the inhibitory population.

[1] E represents the excitatory populations, I stands for the inhibitory populations, the red arrows indicate the excitatory projections and blue arrows show the inhibitory projections, i_I is the external inputs of the I populations, and i_E is the external inputs of the E populations. Li et al. (2022)

0.2.2 Sigmoid Parameters

- a_E, a_I — gain (steepness) of the activation function for E and I populations. Higher gain means a sharper threshold-like response.
- θ_E, θ_I — threshold of the activation function. The input level at which the population reaches half-maximal activation.

0.2.3 Connection Weights

- w_{EE} — excitatory → excitatory coupling strength (recurrent excitation).
- w_{EI} — inhibitory → excitatory coupling strength (how much I suppresses E).
- w_{IE} — excitatory → inhibitory coupling strength (how much E drives I).
- w_{II} — inhibitory → inhibitory coupling strength (recurrent inhibition).

0.2.4 External Inputs

- I_E^{ext} — external input current to the excitatory population (primary control parameter for bifurcation analysis).
- I_I^{ext} — external input current to the inhibitory population (typically fixed at 0).

0.2.5 Parameter Values

Parameter	Value	Description
τ_E	1.0 ms	Average synaptic time constant for excitatory population
τ_I	2.0 ms	Average synaptic time constant for inhibitory population
a_E	1.2	Excitatory gain
a_I	1.0	Inhibitory gain
θ_E	2.8	Excitatory threshold
θ_I	4.0	Inhibitory threshold
w_{EE}	9.0	Strength of self-excitatory feedback
w_{EI}	4.0	Strength of connections between E–I
w_{IE}	13.0	Strength of connections between I–E
w_{II}	11.0	Strength of self-inhibitory feedback
I_E^{ext}	0.0	External input to excitatory population (sweep this)
I_I^{ext}	0.0	External input to inhibitory population

Table 1: Default parameter values from the Neuromatch Academy computational neuroscience curriculum.

Remark: The parameters play an important role in the system, please read the *Periodic solution (Oscillations)* part in details.

1 Nullcline Analysis

1.1 E-nullcline

Let $\dot{r}_E = 0$, then

$$\frac{1}{\tau_E}(-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}})) = 0 \tag{4}$$

$$-r_E + F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = 0 \tag{5}$$

$$F_E(w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}}) = r_E \tag{6}$$

Then we use the inverse of F (namely the logit function):

$$F^{-1}(y; a, \theta) = \theta + \frac{1}{a} \ln \frac{y + c}{1 - y - c}, \quad c = \frac{1}{1 + e^{a\theta}} \tag{7}$$

Applying F_E^{-1} to both sides:

$$F_E^{-1}(r_E) = w_{EE} \cdot r_E - w_{EI} \cdot r_I + I_E^{\text{ext}} \tag{8}$$

Solving for r_I:

$$r_I = \frac{1}{w_{EI}}(w_{EE} \cdot r_E - F_E^{-1}(r_E) + I_E^{\text{ext}}) \tag{9}$$

1.2 I-nullcline

Analogously, let $\dot{r}_I = 0$, we have

$$r_I = F_I(w_{IE} \cdot r_E - w_{II} \cdot r_I + I_I^{\text{ext}}) \tag{10}$$

$$r_E = \frac{1}{w_{IE}}(w_{II} \cdot r_I + F_I^{-1}(r_I) - I_I^{\text{ext}}) \tag{11}$$

1.3 Phase plane

After substituting the parameters, we obtain the figure

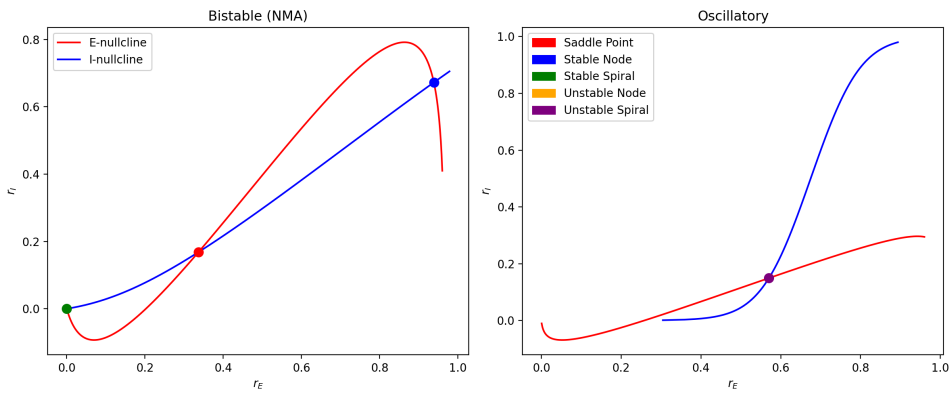

Figure 2: Phase plane analysis

Inserting point $(0.1, 0.5)$, it's easy to check the arrow direction. We omit here.

Remark: The domains are $r_E \in (0, 0.9664)$ and $r_I \in (0, 0.9820)$, as the logarithm requires its argument to be positive.

2 Qualitative Analysis

Note that in the previous section, we obtain the phase plane with 3 equilibrium points. Now we use the Jacobian matrix to determine its state. Recall the Jacobian matrix:

$$J = \begin{pmatrix} \frac{\partial \dot{r}_E}{\partial r_E} & \frac{\partial \dot{r}_E}{\partial r_I} \\ \frac{\partial \dot{r}_I}{\partial r_E} & \frac{\partial \dot{r}_I}{\partial r_I} \end{pmatrix} \tag{12}$$

We solve the partial derivative and obtain

$$J = \begin{pmatrix} \frac{1}{\tau_E}(-1 + w_{EE} \cdot F'_E) & \frac{1}{\tau_E}(-w_{EI} \cdot F'_E) \\ \frac{1}{\tau_I}(w_{IE} \cdot F'_I) & \frac{1}{\tau_I}(-1 - w_{II} \cdot F'_I) \end{pmatrix} \tag{13}$$

in which

$$F'(x) = a \cdot (F(x) + c)(1 - F(x) - c), \quad c = \frac{1}{1 + e^{a\theta}} \tag{14}$$

Note that F' is evaluated at the sigmoid input, not at r_E or r_I directly:

$$u_E = w_{EE} r_E - w_{EI} r_I + I_E^{\text{ext}}, \quad u_I = w_{IE} r_E - w_{II} r_I + I_I^{\text{ext}} \tag{15}$$

We classify each fixed point using the trace $\tau = \text{tr}(J)$, determinant $\Delta = \det(J)$, and discriminant $\tau^2 - 4\Delta$ of the Jacobian:

- $\Delta < 0 \Rightarrow$ saddle point
- $\Delta > 0, \tau < 0 \Rightarrow$ stable (node if $\tau^2 - 4\Delta > 0$, spiral if < 0)
- $\Delta > 0, \tau > 0 \Rightarrow$ unstable

For our 3 fixed points:

Fixed Point	τ	Δ	$\tau^2 - 4\Delta$	**Classification**
$(0, 0)$	-1.247	0.406	-0.069	Stable Spiral
$(\approx 0.34, 0.17)$	—	-0.941	—	Saddle Point
$(\approx 0.94, 0.68)$	-2.403	1.381	0.250	Stable Node

Table 2: Fixed point classification for default parameters ($I_E^{\text{ext}} = 0$).

The system with default parameters ($I_E^{\text{ext}} = 0$) is therefore **bistable**. (In a dynamical system, **bistability** means the system has two stable equilibrium states a bistable structure can be resting in either of two states.)

3 Periodic Solutions (Oscillations)

Recall the **Poincaré-Bendixson theorem** states that for a two-dimensional system, for which we know that the solutions will stay in **a bounded area** after some finite time and

there is **no fixed point**, there must exist a periodic solution to which the trajectories converge.

We sweep the I_E^{ext} until we obtain only one fixed point which is unstable.

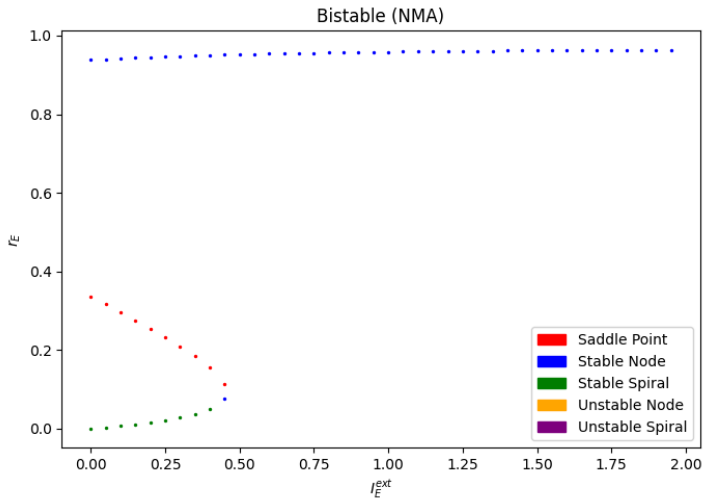

Figure 3: Stable fixed point output after sweeping I_E^{ext}

Note that after sweeping I_E^{ext}, we only get the stable 1 fixed point, why?

Comparing with the parameters used in Li et al. (2022) (Table 1, based on Wilson and Cowan (1972) and Jadi and Sejnowski (2014)) for gamma oscillation research:

Parameter	**NMA (bistability)**	**Table 1 (oscillation)**
w_{EE}	9	16
w_{EI}	4	26
w_{IE}	13	20
w_{II}	11	1
τ_E (ms)	1	20
τ_I (ms)	2	10
θ_E, θ_I	2.8, 4.0	5, 20
$I_E^{\text{ext}}, I_I^{\text{ext}}$	0, 0	2, 7

Table 3: Parameter comparison: bistable vs. oscillatory regime.

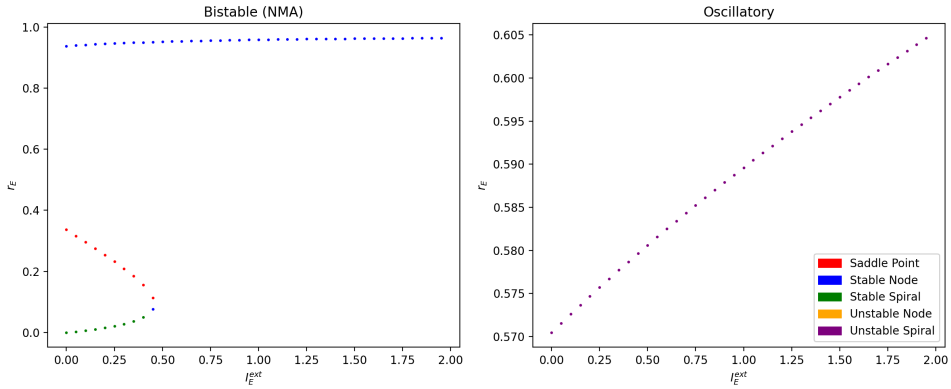

Figure 4: Output comparison between NMA (bistable) and Li et al. (oscillatory) parameter regimes.

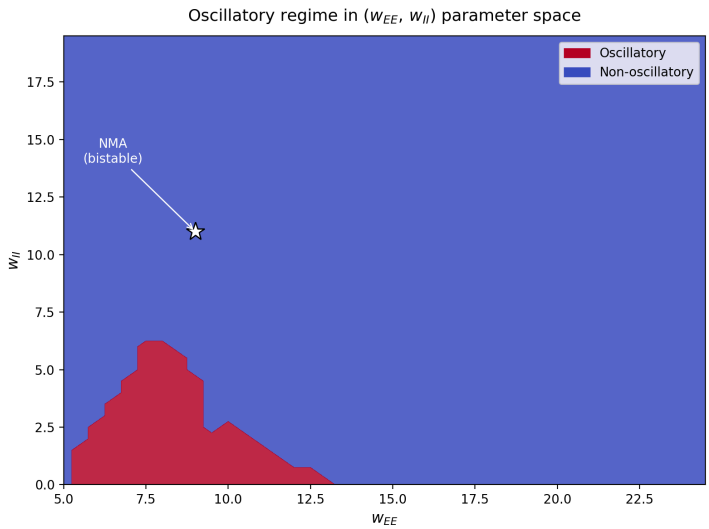

Figure 5: Oscillatory regime in (Wee, WI) parameter space for NMA parameters

3.1 Conditions for Oscillation

A Hopf bifurcation requires a fixed point where the Jacobian has both positive trace (instability) and positive determinant (complex eigenvalues):

$$\tau = \frac{1}{\tau_E}(-1 + w_{EE}F'_E) + \frac{1}{\tau_I}(-1 - w_{II}F'_I) \tag{16}$$

$$\Delta = \frac{1}{\tau_E\tau_I}[1 + w_{II}F'_I - w_{EE}F'_E + (w_{EI}w_{IE} - w_{EE}w_{II})F'_E F'_I] \tag{17}$$

The trace condition $\tau > 0$ requires $w_{EE}F'_E$ to be large: excitatory recurrence must overcome intrinsic decay. In the determinant, $w_{EE}F'_E$ appears in two terms: $-w_{EE}F'_E$ (which pushes Δ down) and the cross-term $(w_{EI}w_{IE} - w_{EE}w_{II})F'_E F'_I$. The sign of $w_{EI}w_{IE} - w_{EE}w_{II}$ determines whether the two conditions cooperate or compete.

This yields three **necessary**[2] conditions for oscillation:

[2]Here we mean necessary condition since it's **not sufficient**. It is crucial to note that while no oscillatory state (Unstable Spiral) can violate these conditions, satisfying them is necessary but not sufficient for a Hopf bifurcation. Even when Condition 2 holds, the positive cross-term $(w_{EI}w_{IE} - w_{EE}w_{II})F'_E F'_{I'}$ must be large enough to overcome the negative trace-related term $-w_{EE}F'_E$ and keep the entire determinant $\Delta > 0$. If this

Condition 1: $w_{EE}F'_E$ must be large enough that the excitatory term overcomes intrinsic decay $w_{EE}F'_E > \tau_E\big(1 + \frac{1}{\tau_I}(1 + w_{II}F'_I)\big)$.

Condition 2: $w_{EI}w_{IE} > w_{EE}w_{II}$. When this holds, increasing F'_E pushes the trace up and helps the determinant stay positive, which means that the two Hopf conditions are cooperative. When violated, increasing F'_E raises the trace but drags the determinant down: no fixed point can satisfy both conditions simultaneously.

Remark: What if $w_{EI}w_{IE} = w_{EE}w_{II}$? Let us assume $w_{EI}w_{IE} = w_{EE}w_{II}$. Then

$$\Delta = \frac{1}{\tau_E\tau_I}(1 + w_{II}F'_I - w_{EE}F'_E) \tag{18}$$

For $\Delta > 0$, we need $1 + w_{II}F'_I > w_{EE}F'_E$. For $\tau > 0$, $w_{EE}F'_E > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F'_I)$. Thus for both hold simultaneously, we need $1 + w_{II}F'_I > w_{EE}F'_E > 1 + \frac{\tau_E}{\tau_I}(1 + w_{II}F'_I)$. Note that it may hold when $\frac{\tau_E}{\tau_I} < 1$, i.e. $\tau_E < \tau_I$.
From Keeley et al. (2019), "weakly modulated network gamma is found when the firing rate of the inhibitory population is recruited faster than that of the excitatory population, and strongly synchronized gamma oscillations are seen for the reverse. In both frameworks, network oscillations arise when a steady state loses stability via Hopf bifurcation as drive to the excitatory population increases. The small-amplitude oscillation that emerges from the bifurcation persists over a substantial range of values of the drive in the weakly modulated regime, but for the strongly synchronous case the oscillation amplitude rises sharply to become a strongly synchronous E-I rhythm as drive increases".
It shows that the Oscillation may occur when $\tau_E < \tau_I$ and $w_{EI}w_{IE} = w_{EE}w_{II}$. But since it is not Gamma oscillation, we don't count the equality. And it implies that $1 + w_{II}F'_I \leq w_{EE}F'_E$, so we need $w_{EI}w_{IE} - w_{EE}w_{II} > 0$ to keep the $\Delta > 0$.

Condition 3: w_{EE} must lie in a finite window. The sigmoid derivative F'_E is maximised when $u_E = w_{EE}r_E - w_{EI}r_I + I_E^{\text{ext}}$ lies near the midpoint θ_E. If w_{EE} is too small, u_E stays below θ_E and F'_E is small. If w_{EE} is too large, strong positive feedback pushes the fixed point past θ_E into saturation, and F'_E is again small. Only intermediate w_{EE} places the fixed point in the steep region of the sigmoid.

3.1.1 Verification[3]

Numerical evaluation at selected points in the (w_{EE}, w_{II}) parameter space confirms all three conditions. At oscillatory points (e.g. $w_{EE} = 6$, $w_{II} = 1$), the system has a single Unstable Spiral with $u_E = 2.27$ (near $\theta_E = 2.8$), $F'_E = 0.27$ (near the maximum $a_E/4 = 0.3$), and a positive cross-term of $w_{EI}w_{IE} - w_{EE}w_{II} = 46$. All three conditions are satisfied.

Points that violate individual conditions fail to oscillate:

- $(w_{EE} = 10.5, w_{II} = 5.0)$: cross-term ≈ 0 — condition 2 fails at the boundary.
- $(w_{EE} = 12, w_{II} = 1)$: cross-term $= 40 > 0$, but w_{EE} too large pushes the fixed point past θ_E into saturation — condition 3 fails.
- $(w_{EE} = 10, w_{II} = 3)$: cross-term $= 22 > 0$, but $w_{EE}F'_E$ insufficient — condition 1 fails.

cooperative "pull" is insufficient, the determinant drops below zero ($\Delta < 0$), and the system inevitably collapses into a Saddle Point, despite satisfying all three necessary conditions.

[3]To verify it, use our condition test module.

All observed oscillations occur through Hopf bifurcation at a single fixed point (Unstable Spiral). No oscillation via the global three-fixed-point mechanism (two Unstable Spirals + Saddle) was found in the parameter range surveyed.

3.1.2 Application to the parameters

For the NMA default parameters ($w_{EE} = 9$, $w_{EI} = 4$, $w_{IE} = 13$, $w_{II} = 11$):

$$w_{EI}w_{IE} = 52 < 99 = w_{EE}w_{II} \tag{19}$$

Condition 2 is violated: the cross-term is $-47F_E'F_I'$, strongly competitive. The trace and determinant conditions cannot be simultaneously satisfied at any fixed point. Numerically, the saddle point achieves F_E' up to 0.28 with positive trace, but $\Delta < 0$; the stable spiral has $\Delta > 0$, but $F_E' \leq 0.12$ and negative trace. No Hopf bifurcation occurs regardless of I_E^{ext}.

For Li et al. ($w_{EE} = 16$, $w_{EI} = 26$, $w_{IE} = 20$, $w_{II} = 1$, $\theta_E = 5.0$): $w_{EI}w_{IE} = 520 \gg 16 = w_{EE}w_{II}$ (condition 2), $u_E = 5.24 \approx \theta_E = 5.0$ (condition 3), and $w_{EE}F_E' = 4.7$ (condition 1). All three conditions are satisfied, and the system exhibits a Hopf bifurcation at a single Unstable Spiral.

3.1.3 Further discussion

Our analytical conditions are consistent with the numerical bifurcation results of Li et al. (2022), who investigated the role of self-feedback loops in regulating gamma oscillations in the Wilson-Cowan model. Their one-parameter bifurcation with respect to w_{II} (their Figure 3A) shows that increasing inhibitory self-feedback suppresses oscillation, which directly follows from our Condition 2: since $w_{EI}w_{IE}$ is fixed, increasing w_{II} raises $w_{EE}w_{II}$ and eventually violates the cross-term inequality.
For excitatory self-feedback, their one-parameter bifurcation with respect to w_{EE} (their Figure 6A) reveals that oscillation exists only within a finite window $13.57 < w_{EE} < 35$, confirming our Condition 3. However, their summary conclusion that "excitatory self-feedback promotes the generation of gamma oscillations" captures only the lower boundary of this window. Our analysis clarifies why both boundaries exist: the lower bound reflects Condition 1 (insufficient excitatory gain to destabilize the fixed point), while the upper bound reflects Condition 3 (excessive recurrence drives the fixed point past the sigmoid's sensitive region into saturation, reducing F_E'). Their transfer function analysis (their Eq. 13–14) assumes $w_{EE}K_E < 1$, which corresponds precisely to the interior of this window.

We refer to the following papers:

Inhibitory self-feedback. Li et al. (2022):

> The present results reveal that, on one hand, the inhibitory self-feedback loop is not conducive to the generation of gamma oscillations, and increased inhibitory self-feedback strength facilitates the enhancement of the oscillation frequency. On the other hand, the excitatory self-feedback loop promotes the generation of gamma oscillations [...] Inhibitory and excitatory self-feedback loops play a complementary role in generating and regulating the gamma oscillation in Wilson-Cowan model.

Time constants. Keeley et al. (2019):

The models' flexibility to capture the broad range of gamma behavior depends directly on the timescales that represent recruitment of the excitatory and inhibitory firing rates. In particular, we find that weakly modulated gamma oscillations occur robustly when the recruitment timescale of inhibition is faster than that of excitation.

4 Key References

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7. **Neuromatch Academy — Computational Neuroscience, W2D4: Dynamic Networks**